

Communication Protocol for SRNE Street Lamp

Version	Content Modified	Date	Modified by
5.0	New edition of MODBUS protocol, compatible with household system	2019-6-1	Liang Yaohui
5.1	1. Add 485 device address modification command. 2. Add MPL product manual switch load command. 3. Add power disconnect command of external communication module.	2021-4-15	Liang Yaohui
5.2	Correct the line sequence of "1 485 interface description and serial port configuration information"; Increase the page number and make appropriate typesetting.	2024-10-24	Liang Yaohui

Tel: 0755-26458295

Fax: 0755-27325578

Website: www.srnesolar.com

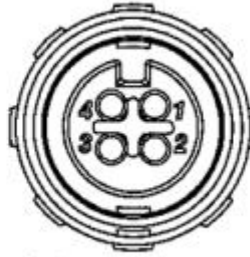
TABLE OF CONTENTS

1 CONFIGURATION INFORMATION OF 485 INTERFACE.....	- 3 -
2 FORMAT AND INSTRUCTION OF COMMUNICATION PROTOCOL.....	- 3 -
2.1 FORMAT.....	- 3 -
2.2 INTRODUCTION.....	- 3 -
2.3 EXAMPLE.....	- 4 -
3 ADDRESS OF DATA PDU.....	- 9 -
3.1 PARTITION SPECIFICATION OF PDU ADDRESS.....	- 9 -
3.2 P0 AREA- PRODUCT INFORMATION AREA.....	- 11 -
3.3 P01 AREA – REAL-TIME DATA AREA OF CONTROLLER.....	- 13 -
3.4 P03 AREA- DEVICE CONTROL COMMAND AREA.....	- 24 -
3.5 P05 AREA - PARAMETER SETTING AREA OF BATTERY.....	- 28 -
3.6 P07 AREA- STREET LAMP LOAD PARAMETER SETTING AREA.....	- 38 -
4 DESCRIPTION OF INFORMATION READING.....	- 46 -
4.1 COMMAND PARSING AND EXAMPLE OF SYSTEM INFORMATION (P0).....	- 46 -
4.2 COMMAND PARSING AND EXAMPLE OF SYSTEM REAL-TIME DATA (P1).....	- 48 -
5 INSTRUCTION OF CONTROL COMMAND (P3 AREA).....	- 57 -
5.1 CONSTANT CURRENT SERIES MANUAL ON/OFF LIGHTS WITH DIMMING SETTINGS.....	- 57 -
5.2 VOLTAGE-REGULATED LOAD SERIES CONTROLLERS TURN ON/OFF THE LOAD.....	- 57 -
5.3 RESTE CONTROL OF CONTROLLER.....	-58-
5.4 COMMUNICATION MODULE POWER-OFF COMMAND.....	-58-

6 SETTING PARAMETER DESCRIPTION OF ACCUMULATOR (P5 AREA)	- 59 -
6.1 PARAMETER SETTING OF LEAD ACID BATTERY	- 59 -
6.2 PARAMETER SETTING REFERENCE OF LITHIUM BATTERY	- 61 -
7 DESCRIPTION OF LOAD SETTING PARAMETER (P7 AREA)	- 64 -
7.1 SETTING OF LOAD CURRENTENT	- 64 -
7.2 SETTING OF INTELLIGENT POWER	- 64 -
7.3 LIGHT CONTROL DELAY SETTING	- 65 -
7.4 SENSING DELAY SETTING	- 65 -
7.5 SETTING OF OPERATION DURATION AND POWER	- 65 -

1 Configuration information of 485 interface

1. Yellow A/D⁺/TX
2. White B/D⁻/RX
3. Black GND
4. Red VCC



Serial port baud rate: 9600 Check bit:NONE Data bit:8bit Stop bit:1bit

2 Format and Instruction of Communication Protocol

2.1 Format

Start Character	Address Field	Function Code	Data	Error Check	Ending Character
>10ms	01H ~ F7H	03H/06H/10H	N	CRC	>10ms
	1 byte	1 byte	N bytes	2 bytes	

2.2 Introduction

The following data with "H" are hexadecimal data and others are decimal data.

- 1) Start character: > 10ms
- 2) Address field: 1 byte, range: 01H ~ F7H (decimal 1 ~ 247), 00h is the broadcast address, all slaves respond but do not return the command
- 3) Function code: 1 byte

Command Name	Type of Accessing Data	Function Code	Error Code
Read single-word or multiple-word	2 bytes	03H	83H

registers			
Write single-word or multiple-word registers	2 bytes	06H	86H
Write continuous N registers	2 bytes	10H	90H
Restore factory default			
Clear historical record			

4) Data: N bytes

5) Error check: 2 bytes, CRC check sum for different bytes of device address, function code and data

6) End mark:>10ms

Note:

1) Data address and data are 2 bytes, the high byte is sent first, then the low byte is sent; however, regarding CRC, the low byte is sent first, then the high byte is sent.

2) There is error in the frame data issued by the error code to the server; the error exception response function code is returned from Client; error code = function code | 80H

2.3 Example

2.4.1 Read register

Request:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	03H
Start address	WORD	0000H ~ FFFFH
Word count read	WORD	0001H ~ 007DH
Check code	WORD	CRC check sum of all above bytes

Normal response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	03H
Data length	BYTE	01H ~ FAH
Data content	WORD	Read data (issue high bit and then low bit)
...	WORD	Read data (issue high bit and then low bit)
Check code	WORD	CRC check sum of all above bytes

Exception response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Error code	BYTE	83H
Exception code	BYTE	N (N=1 , 2 , 3 , 4)
Check code	WORD	CRC check sum of all above bytes

2.4.2 Write single register**Request:**

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	06H
Start address	WORD	0000H ~ FFFFH
Data written	WORD	0000H ~ FFFFH
Check code	WORD	CRC check sum of all above bytes

Normal response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	06H
Start address	WORD	0000H ~ FFFFH
Word count written	WORD	0000H ~ FFFFH
Check code	WORD	CRC check sum of all above bytes

Exception response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Error code	BYTE	86H
Exception code	BYTE	N (N=1 , 2 , 3 , 4)
Check code	WORD	CRC check sum of all above bytes

2.4.3 Write N continuous register**Request:**

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	10H
Start address	WORD	0000H ~ FFFFH
Count written	WORD	0001H ~ 007DH
Quantity of byte written	BYTE	Double of word count
Data content	WORD	Written data (issue high bit and then lower bit)
...	WORD	Written data (issue high bit and then lower bit)

Check code	WORD	CRC check sum of all above bytes
------------	------	----------------------------------

Normal response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Function code	BYTE	10H
Start address	WORD	0000H ~ FFFFH
Count written	WORD	0001H ~ 007DH
Check code	WORD	CRC check sum of all above bytes

Exception response:

Description	Number of Bytes	Command
Device address	BYTE	01H ~ F7H
Error code	BYTE	90H
Exception code	BYTE	N (N=1 , 2 , 3 , 4)
Check code	WORD	CRC check sum of all above bytes

3 Address of Data PDU

3.1 Partition specification of PDU address

Parti tion	Address Distribution	Space Size (word)	Functional Specification	Remark
P0	0x000A ~ 0x001A	17	System information of device	Data of this partition may not be read
P1	0x00FD ~ 0x012D	49	Real-time data of system	1. It is recommended to only simply use address values 0xFD ~ 0x109, 13 words and 26 bytes in total. 2. Other address data can be selectively read in accordance with practical demands, but not fully used. 3. 0xFD and 0x120 are backup registers mutually, 0xFE~0xFF and 0x121~0x122 are backup registers mutually. Data can be accessed through any group of addresses.
P2	Retained			

P3	0xDF00 ~ 0xDF0D	14	Device control zone	Only use manual light on and off command DF0A: power for manual on and off DF0B: time for manual on and off
P4	Retained			
P5	0xE000 ~ 0xE01F	32	Parameter setting of battery	Distinguish parameter setting value of lead acid battery and lithium battery.
P6	Retained			
P7	0xE08D ~ 0xE0B3	39	Parameter setting of street lamp load	0xE092~0xE0AF 10time intervals may not be fully used. Data of time intervals not used are all set as 0.

3.2 P0 area- product information area

Address HEX	Length	Name	English Name	Read and Write	Multiplying power	Unit	Display Format	With or without symbol	Minimum Value	Maximum Value	Default	Remark
A	1	Maximal support voltage of system and rated charge Currentent	Max Volt And charge Rage	R	1	-	%d	N/A				High 8 bits: system voltage Low 8 bits: rated charge Currentent 0CH (decimal 12) 0AH (decimal 10) 18H(decimal 24) 14H(decimal 20) 24H(decimal 36) 1EH(decimal 30) 30H(decimal 48) 2DH(decimal 45) 60H(decimal 96) 3CH(decimal 60) FFH(decimal 255) automatic identification

B	1	Rated discharge Currentent and product type	discharge Rate And Mach Type	R	1	-	%d	N/A				High 8 bits: rated discharge Currentent 0AH (decimal 10) 14H (decimal 20) 1EH(decimal 30) 2DH(decimal 45) 3CH (decimal 60) Low 8 bits: product type 00(controller, for household) 01(controller, street lamp) 03(inverter) 04(integrator of controller and inverter)
C	8	Product Specification	Product Type Str	R	1	-	%s	N/A				New product: specification information MPPT 2440N10-20
14	2	Software	Software	R	1	-	%x	N/A				

		version	Version									
16	2	Hardware version	Hardware Version	R	1	-	%x	N/A				
18	2	Serial number of product	Product SN	R	1	-	%x	N/A				
1A	1	Device (controller) address	Rs485Address	RW	1	-	%d	N/A	1	247	1	

3.3 P01 area – Real-time data area of controller

Address HEX	Length	Name	English Name	Read and Write	Multiplying	power	Unit	Display Format	With or without symbol	Minimum Value	Maximum Value	Default	Remark
FD	1	Load state	Load And	R	1	-	%d	N/A					High 8 bits:

		and charge state 0	Charge State 0									b7: load state (0: load off, 1: load on) b0 ~ b6: intensity value (0% ~100%) Low 8 bits: 00H: no charge is started 01H: charge mode is started 02H: mppt charge mode 03H: equalizing charge mode 04H: boost charge mode 05H: float charge mode 06H: Currentent restriction (super power)
FE	2	Controller failure and alarm	FaultInfoL ed0	R	1	-	%d	N/A				Details are shown in the instruction

		information 0										
100	1	Battery capacity SOC	Bat Soc	R	1	-	%d	N/A	0	100		
101	1	Battery voltage	Bat Volt	R	0.1	V	%.1fV	N/A				
102	1	Charge Currentent	Charge Currentent	R	0.01	A	%.2fA	N/A				
103	1	Device temperature (controller) and battery temperature	Device Bat Temper	R	1	°C	%d°C	N/A				High 8 bits: controller temperature Low 8 bits: battery temperature b7: symbol bit (0 indicates + and 1 indicates -) b0-b6: temperature value
104	1	DC load	Load Dc	R	0.1	V	%.1fV	N/A				

		voltage	Volt									
105	1	DC load Currentent	Load Dc Currentent	R	0.01	A	%.2fA	N/A				
106	1	DC load power	Load Dc Power	R	1	W	%dW	N/A				
107	1	Voltage of solar panel	Pv Volt	R	0.1	V	%.1fV	N/A				
108	1	Currentent of solar panel	Pv Currentent	R	0.01	A	%.2fA	N/A				
109	1	charge power	Charge Power	R	1	W	%dW	N/A				
10A	1	DC load on/off command	Dc Load Switch Command	W	1	-	%d	N/A				Street lamp product is not achieved

10B	1	Minimum battery voltage on the day	Bat Min Volt Today	R	0.1	V	%.1fV	N/A				Historical data
10C	1	Maximum battery voltage on the day	Bat Max Volt Today	R	0.1	V	%.1fV	N/A				
10D	1	Maximum charge Currentent on the day	Bat Max Charge Currentent Today	R	0.01	A	%.2fA	N/A				
10E	1	Maximum discharge	Bat Max Discharge	R	0.01	A	%.2fA	N/A				

		Currentent on the day	Currentent Today									
10F	1	Maximum charge power on the day	Bat Max Charge Power Today	R	1	W	%dW	N/A				
110	1	Maximum discharge power on the day	Bat Max Discharge Power Today	R	1	W	%dW	N/A				
111	1	Charge ampere hour on the day	Bat Charge AH Today	R	1	AH	%dAH	N/A				
112	1	Discharge	Bat	R	1	AH	%dAH	N/A				

		ampere hour on the day	Discharge AH Today									
113	1	Generating capacity on the day	Generatin g Energy Today	R	1	Wh	%dWh	N/A				
114	1	Electricity consumption on the day	Used Energy Today	R	1	Wh	%dWh	N/A				
115	1	Total operating days	Work Days Total	R	1	天	%d天	N/A				
116	1	Total over-dischar ge times of	Bat Over-Disc harge	R	1	-	%d	N/A				

		battery	Times									
117	1	Total charge times of battery	Bat Over-Charge Times	R	1	-	%d	N/A				
118	2	Total charge ampere hour of battery	Bat Charge AH Total	R	1	AH	%dAH	N/A				
11A	2	Total discharge ampere hour of battery	Bat Discharge AH Total	R	1	AH	%dAH	N/A				
11C	2	Accumulated generating capacity	Generating Energy Total	R	1	Wh	%dWh	N/A				

11E	2	Accumulated electricity consumption	Used Energy Total	R	1	Wh	%dWh	N/A				
120	1	Load state and charge state 1	LoadAndC hargeState 1	R	1	-	%d	N/A				The data are the same as the data of register 0xFD, the specific analyses are shown in the register: 0xFD
121	2	Controller failure, alarm information 1	FaultInfoL ed1	R	1	-	%d	N/A				The data are the same as registers 0xFE and 0xFF, specific analyses are shown in registers : 0xFE and 0xFF
123	1	Maximum battery temperature on the day	Bat Highest Temper	R	1	°C	%d°C	N/A				

124	1	Minimum battery temperature on the day	Bat Lowest Temper	R	1	°C	%d°C	N/A				
125	2	Total operation duration of load (accumulate d lamp on)	Load Total Work Time	R	1	S	%dS	N/A				
127	1	On duration on the day (attended)	Led Sensor On Time	R	1	S	%dS	N/A				In case of senseless mode, attended duration is the on time on the day.
128	1	On duration	Led Sensor	R	1	S	%dS	N/A				

		on the day (unattended)	Off Time									
129	1	Lighting index	Led Light On Index	R	1	-	%d	N/A				
12A	1	Energy consumption index	Power Save Index	R	1	-	%d	N/A				
12B	1	Health index	Sys Health Index	R	1	-	%d	N/A				
12C	1	Charge duration on the day	Bat Charge Time	R	1	S	%dS	N/A				
12D	1	Night duration	Night Length	R	1	S	%dS	N/A				

3.4 P03 area- device control command area

Address HEX	Length h	Name	English Name	Read and Write	Multiplying power	Unit	Display Format	With or without symbol	Minimum Value	Maximum Value	Default	Remark
DF00	1	On and off control	Command Power On Off	RW	1	-	%x	N/A				0: off, no charging or discharging 1: on Others: no action (not achieved)
DF01	1	Reset control	Command Machine Reset	W	1	-	%x	N/A				1: reset Others: inaction
DF02	1	Restore factory	Command Restore	W	1	-	%x	N/A				1: restoration Others: inaction

		defaults	Factory Setting									Restore factory defaults and clear all accumulated information, restore the parameter to the default state (not achieved)
DF03	1	Clear Currentent alarm	Command Clear Alarm	W	1	-	%x	N/A				1: clear Others: inaction (not achieved)
DF04	1	Clear statistics	Command Clear Statistic	W	1	-	%x	N/A				1: clear Others: inaction (not achieved)
DF05	1	Clear historical record	Command Clear His Record	W	1	-	%x	N/A				1: clear Others: inaction (not achieved)

DF06	2	Firmware upgrade command	Command Upgrade Command	W	1	-	%x	N/A				Firmware upgrade command (not achieved)
DF08	1	Dormancy control/activation command	Command Sleep Run	W	1	-	%x	N/A				5A5A:sleep A5A5:run (not achieved)
DF09	1	Load operation mode command	Command Load Mode	W	1	-	%x	N/A				High 8 bits: 00: automatic mode 01: test mode 02: manual mode 03: timing mode Low 8 bits: bit 0: 1 lamp on at manual

												mode; 0 lamp off at manual mode (not achieved)
DF0A	1	Manual on power	Command Test Power	W	1	W	%d%	N/A	0%	100%		On power at test mode 0~100%
DF0B	1	Manual on duration	Command Test Time	W	1	S	%dS	N/A	5S	54000S		On duration at test mode 0~54000S
DF0C	1	Commercial power switching command	Command Switch Dc	W	1	-	%x	N/A				1: switch to mains power supply 0: power supply from battery
DF0D	1	Immediate equalizing	Command Bat Equal	RW	1		%d	N/A	0	1	1	0: disabled 1: enabled (not achieved)

		charge	Charge									
		command	Immediate									

3.5 P05 area - parameter setting area of battery

Address HEX	Length	Name	English Name	Read and Write	Multiplying power	Unit	Display Format	With or without symbol	Minimum Value	Maximum Value	Default	Remark
E001	1	Photovoltaic charging Currentent setting	Pv Charge Currentent Set	RW	0.01	A	%2fA	No	0.00A	600.00A	20.00A	Not achieved
E002	1	Nominal capacity of	Bat Rate Cap	RW	1	AH	%dAH	No	0AH	9999AH	200AH	Not achieved

		battery										
E003	1	System voltage setting	Bat Rate Volt	RW	1	V	%dV	No	12V	255V	12V	03: 3V 06: 6V 12: 12V 24: 24V 36: 36V 48: 48V FF: automatic identification Others: automatic identification
E004	1	Battery type	Bat Type Set	RW	1	-	%d	No	0	11	10	10= User-defined (lead-acid battery) 11= User-defined

												(lithium battery)
E005	1	Over voltage	Bat Over Volt	RW	0.1	V	%1fV	No	1.00V	17.0V	15.5V	
E006	1	Charging limit voltage	Bat Charge Limit Volt	RW	0.1	V	%1fV	No	1.00V	17.0V	15.0V	
E007	1	Equalizing charging voltage	Bat Const Charge Volt	RW	0.1	V	%1fV	No	1.00V	17.00V	14.6V	
E008	1	Boost charging voltage/Overcharge voltage of lithium battery	Bat Improve Charge Volt	RW	0.1	V	%1fV	No	1.00V	17.00V	14.4V	
E009	1	Floating charge voltage/Lithium	Bat Float Charge	RW	0.1	V	%1fV	No	1.00V	17.00V	13.8V	

		overcharge return voltage	Volt									
E00A	1	Boost charging return voltage	Bat Improve Charge Back Volt	RW	0.1	V	%.1fV	No	1.00V	17.00V	13.2V	
E00B	1	Over-discharge return voltage	Bat Over-disch arge Back Volt	RW	0.1	V	%.1fV	No	1.00V	17.00V	12.6V	
E00C	1	Under voltage warning voltage	Bat Under Volt	RW	0.1	V	%.1fV	No	1.00V	17.00V	12.0V	Street lamp product is not achieved
E00D	1	Over-discharge voltage	Bat Over-disch	RW	0.1	V	%.1fV	No	1.00V	17.00V	11.0V	

			arge Volt									
E00E	1	Discharge limiting voltage	Bat Discharge Limit Volt	RW	0.1	V	%.1fV	No	1.00V	17.00V	10.6V	Street lamp product is not achieved
E00F	1	Charge cut-off SOC, discharge cut-off SOC	Bat Stop SOC	RW	1	-	%d%	No	0	65535	25610	High 8 bits: charge cut-off SOC Low 8 bits: discharge cut-off SOC street lamp product is not achieved
E010	1	Over-discharge delay duration	Bat Over-discharge Delay Time	RW	1	S	%dS	No	0S	120S	5S	Street lamp product is not achieved
E011	1	Equalizing	Bat Const	RW	1	Min	%dmin	No	0min	600min	120min	

		charge duration	Charge Time									
E012	1	Boost charge duration	Bat Improve Charge Time	RW	1	Min	%dmin	No	10min	600min	120min	
E013	1	Equalizing charge interval	Bat Const Charge Gap Time	RW	1	day	%d 天	No	0 day	255 days	30 days	
E014	1	Temperature compensation coefficient	Coeff Temper Compen	RW	1	mV/ °C /2V	%d	Yes	0	10	5	Special for lead acid, unavailable for lithium battery
E015	1	Maximum battery	Charge Max	RW	1	°C	%d°C	Yes	-40°C	120°C	60°C	High 8 bits not used Low 8 bits: temperature

		charging temperature	Temper									value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value
E016	1	Minimum battery charging temperature	Charge Min Temper	RW	1	°C	%d°C	Yes	-40°C	120°C	-30°C	High 8 bits are not used Low 8 bits: temperature value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value
E017	1	Maximum battery	Discharge Max	RW	1	°C	%d°C	Yes	-40°C	120°C	60°C	High 8 bits are not used Low 8 bits: temperature

		discharging temperature	Temper									value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value
E018	1	Minimum battery discharging temperature	Discharge Min Temper	RW	1	°C	%d°C	Yes	-40°C	120°C	-30°C	High 8 bits are not used Low 8 bits: temperature value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value
E019	1	(Lithium	Heat Bat Start	RW	1	°C	%d°C	Yes	-40°C	120°C	-10°C	High 8 bits are not used Low 8 bits: temperature

		battery) heating start battery temperature	Temper									value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value Street lamp product is not achieved
E01A	1	(Lithium battery) heating stop battery temperature	Heat Bat Stop Temper	RW	1	°C	%d°C	Yes	-40°C	120°C	-5°C	High 8 bits are not used Low 8 bits: temperature value b7: sign bit (0 indicates +, 1 indicates -) b0-b6: temperature value

												Street lamp product is not achieved
E01B	1	Mains switching voltage	Bat Switch Dc Volt	RW	0.1	V	%.1fV	No	1.00V	17.0V	11.5V	
E01C	1	(Lithium battery) Stop charging Currentent	Stop Charge Currentent Set	RW	0.01	A	%.2fA	No	0.00A	600.00A	0.00A	Street lamp product is not achieved
E01D	1	DC load operation mode	Dc Load Mode	RW	1	-	%d	No	0	17	0	Street lamp product is not achieved
E01E	1	Optical control delay duration (household: minute)	Light Control Delay Time	RW	1	Min	%dmin	No	0min	60min	0	Street lamp product is not achieved

E01F	1	Light-control voltage	Light Control Volt	RW	1	V	%dV	No	3V	40V	5V	
------	---	--------------------------	--------------------------	----	---	---	-----	----	----	-----	----	--

3.6 P07 area- street lamp load parameter setting area

Address HEX	Length	Name	English Name	Read and Write	Multiplying power	Unit	Display Format	With or without symbol	Minimum Value	Maximum Value	Default	Remark
E08D	1	Load Currentent setting	Led Current Set	RW	0.01	A	%.2fA	No	0.15A	10.00A	0.90A	
E08E	1	Intelligent power	Auto Power Set	RW	1	-	%d	No	0	6	2	Off/low/middle/high/au tomatic (automatic setting through controller)/user 1 (set by

												user)
E08F	1	Light-control delay duration (S)	Light Control Delay Time S	RW	1	S	%dS	No	0S	3600S	20S	Special for lamp street, unit: s
E090	1	Sensing delay	Sensor Delay Time	RW	1	S	%dS	No	0S	3600S	20S	
E091	1	Sensing distance	Sensor Distance	RW	1	-	%dM	No	0M	20M	10M	Not achieved
E092	1	On duration of first section	Load Time1	RW	1	S	%dS	No	0S	54000S	1800S	30 minutes
E093	1	Attended power of 1 st section	Load Sensor On Power1	RW	1	%	%d%	No	0%	100%	50%	Attended power 50%
E094	1	Unattended power of 1 st	Load Sensor Off Power1	RW	1	%	%d%	No	0%	100%	20%	Unattended power 20%

		section										
E095	1	On duration of 2 nd section	Load Time2	RW	1	S	%dS	No	0S	54000S	1800S	30 minutes
E096	1	Attended power of 2 nd section	Load Sensor On Power2	RW	1	%	%d%	No	0%	100%	70%	Attended power 70%
E097	1	Unattended power of 2 nd section	Load Sensor Off Power2	RW	1	%	%d%	No	0%	100%	30%	Unattended power 30%
E098	1	On duration of 3 rd section	Load Time3	RW	1	S	%dS	No	0S	54000S	7200S	2 hours
E099	1	Attended power of 3 rd section	Load Sensor On Power3	RW	1	%	%d%	No	0%	100%	100%	Attended power 100%
E09A	1	Unattended power of 3 rd	Load Sensor Off Power3	RW	1	%	%d%	No	0%	100%	50%	Unattended power 50%

		section										
E09B	1	On duration of 4 th section	Load Time4	RW	1	S	%dS	No	0S	54000S	1800S	30 minutes
E09C	1	Attended power of 4 th section	Load Sensor On Power4	RW	1	%	%d%	No	0%	100%	70%	Attended power 70%
E09D	1	Unattended power of 4 th section	Load Sensor Off Power4	RW	1	%	%d%	No	0%	100%	30%	Unattended power 30%
E09E	1	On duration of 5 th section	Load Time5	RW	1	S	%dS	No	0S	54000S	1800S	30 minutes
E09F	1	Attended power of 5 th section	Load Sensor On Power5	RW	1	%	%d%	No	0%	100%	50%	Attended power 50%
E0A0	1	Unattended power of 5 th section	Load Sensor Off Power5	RW	1	%	%d%	No	0%	100%	20%	Unattended power 20%

		section										
E0A1	1	On duration of 6 th section	Load Time6	RW	1	S	%dS	No	0S	54000	14400S	4 hours
E0A2	1	Attended power of 6 th section	Load Sensor On Power6	RW	1	%	%d%	No	0%	100%	30%	Attended power 30%
E0A3	1	Unattended power of 6 th section	Load Sensor Off Power6	RW	1	%	%d%	No	0%	100%	10%	Unattended power10%
E0A4	1	On duration of 7 th section	Load Time7	RW	1	S	%dS	No	0S	54000	0	
E0A5	1	Attended power of 7 th section	Load Sensor On Power7	RW	1	%	%d%	No	0%	100%	100%	
E0A6	1	Unattended power of 7 th section	Load Sensor Off Power7	RW	1	%	%d%	No	0%	100%	50%	

		section										
E0A7	1	On duration of 8 th section	Load Time8	RW	1	S	%dS	No	0S	54000	0	
E0A8	1	Attended power of 8 th section	Load Sensor On Power8	RW	1	%	%d%	No	0%	100%	100%	
E0A9	1	Unattended power of 8 th section	Load Sensor Off Power8	RW	1	%	%d%	No	0%	100%	50%	
E0AA	1	On duration of 9 th section	Load Time9	RW	1	S	%dS	No	0S	54000	0	
E0AB	1	Attended power of 9 th section	Load Sensor On Power9	RW	1	%	%d%	No	0%	100%	100%	
E0AC	1	Unattended power of 9 th	Load Sensor Off Power9	RW	1	%	%d%	No	0%	100%	50%	

		section										
E0AD	1	On duration in Morning	Load Morning Light Time	RW	1	S	%dS	No	0S	54000	0	
E0AE	1	Attended power for Morning on	Load Morning Light Sensor On Power	RW	1	%	%d%	No	0%	100%	50%	
E0AF	1	Unattended power for Morning on	Load Morning Light Sensor Off Power	RW	1	%	%d%	No	0%	100%	20%	
E0B0	1	USER1 start voltage of intelligent power	AutoPowerStartUser1Volt	RW	0.1	V	%.1fV	No	9.0V	17.0V	11.5V	
E0B1	1	USER1 final voltage of	AutoPowerStopUser1Volt	RW	0.1	V	%.1fV	No	9.0V	17.0V	10.0V	

		intelligent power										
E0B2	1	USER1 start power of intelligent power	Auto Power Start User1 Power	RW	1	%	%d%	No	1%	100%	100%	Not achieved
E0B3	1	USER1 final power of intelligent power	Auto Power Stop User1 Power	RW	0.01	A	%.2fA	No	0.01A	10.00A	0.1A	

More detailed PUD definitions are shown in *ModBus Register Definition V3*

4 Description of Information Reading

4.1 Command parsing and example of system information (PO)

4.1.1 Read system voltage and Currentent of controller

PDU address	Byte	Read/write	Data		Significance
000AH	2	R	High 8 bits: system voltage	0CH (decimalism 12)	12V
				18H (decimalism 24)	24V
				24H (decimalism 36)	36V
				30H (decimalism 48)	48V
				60H (decimalism 96)	96V
				FFH (decimalism 255)	Automatic identification of system voltage
			Low 8 bits: system Currentent	0AH (decimalism 10)	10A
				1FH	15A

				(decimalism 20)	
				14H (decimalism 20)	20A
				1EH (decimalism 30)	30A
				3CH (decimalism 60)	60A

It is learned from "table 1" that PDU address is 000AH, read a word (two bytes)

Send: 01 03 000A 0001 A408

Receive: 01 03 02 1814 324C

Parsing: high byte 18H indicates that system voltage of the controller is 24V, low byte 14H indicates that system Currentent is 20A.

4.1.2 Read product type of controller

It is learned that PDU addresses are 000CH~0013H in turn, including 16 bytes in total, let the data addresses are for store data (ASCII): ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' ', ' '.

Send : 01 03 000C 0008 840F

Receive : 01 03 10 2020 2020 4D54 3438 3330 2020 2020 2020 EE98

Parsing: product type of the controller: MT4830 (corresponding ASCII of 20H is ' ', blank data, negligible)

4.1.3 Read software version and hardware version of controller

It is learned that PDU addresses are 0014H, 0015H, 0016H and 0017H in turn

Send : 01 03 0014 0004 040D

Receive : 01 03 08 0003 0201 0001 0203 8A54

Parsing: (the highest byte 00H is not used) 030201H indicates that software version of the controller is V03.02.01

(the highest byte 00H is not used) 010203H indicates that hardware version of the controller is V01.02.03

4.1.4 Read product serial number of controller

It is learned that PDU addresses are 0018H and 0019H in turn.

Send : 01 03 0018 0002 740F

Receive : 01 03 04 1501 FFFF AE4F

Parsing: 1501FFFFH refers to product serial number, indicating the 65535th product manufactured in January 2015 (hexadecimal is FFFFH)

4.2 Command parsing and example of system real-time data (P1)

4.2.1 Read battery capacity SOC

It is learned that PDU address is 0100H

Send : 01 03 0100 0001 85F6

Receive : 01 03 02 0064 B9AF

Parsing: (the highest byte 00H is not used) battery capacity SOC is 64H% (decimalism 100%))

4.2.2 Read battery voltage

Voltage ratio of battery is 0.1 and the result is the read value multiplying 0.1, the same below, omitted.

It is learned that PDU address is 0101H

Send : 01 03 0101 0001 D436

Receive : 01 03 02 007B F867

Parsing: *formula(battery voltage = battery voltage* 0.1)*

Battery voltage: (007BH, decimalism 123), 007BH * 0.1 = 12.3 (V)

4.2.3 Read surface temperature of battery and controller temperature

It is learned that PDU address is 0103H

Send : 01 03 0103 0001 75F6

Receive : 01 03 02 2820 A79C

Parsing: 28H indicates the controller temperature is 40°C, if the datum is A8H, it indicates that the controller temperature is -40°C.

20H indicates that surface temperature of battery is 30°C. If the datum is A0H, it indicates that the controller temperature is -30°C.

4.2.4 Read voltage, Currentent and power of street lamp

It is learned that PDU addresses are 0104H, 0105H and 0106H in turn

Send : 01 03 0104 0003 45F6

Receive : 01 03 06 0078 00C8 00F0 00C5

Parsing:

Formula: voltage of street lamp = voltage of street lamp * 0.1

0078H is voltage of street lamp, then voltage of street lamp: $0078H * 0.1 = 120 * 0.1 = 12.0(V)$

Formula: Currentent of street lamp = Currentent of street lamp * 0.01

00C8H is Currentent of street lamp, then Currentent of street lamp: $00C8H * 0.01 = 200 * 0.01 = 2.00 (A)$

00F0H is power of street lamp, decimalism is 240(W); it can be acquired through (voltage of streetlamp * Currentent of street lamp)

4.2.5 Read voltage, charge Currentent and charge power of solar panel

It is learned that PDU address are 0107H, 0108H and 0109H

Send : 01 03 0107 0003 B5F6

Receive : 01 03 06 0090 0096 00D8 011E

Parsing:

Formula: Voltage of solar panel = voltage of solar panel * 0.1

00AAH is voltage of solar panel, then voltage of solar panel: $0090H * 0.1 = 144 * 0.1 = 14.4(V)$

Formula: charge Currentent of solar panel = charge Currentent of solar panel * 0.01

0096H is the charge Currentent of solar panel, then charge Currentent of solar panel: $0096H * 0.01 = 150 * 0.01 = 1.50 (A)$

00D8H is the charge power of solar panel, decimalism is 216(W), it can also be acquired through (voltage of solar panel * charge Currentent of solar panel)

4.2.6 Read historical data on the day

Read following data on the day: minimum battery voltage on the day, maximum battery voltage on the day, maximum charge Currentent on the day, maximum discharge Currentent on the day, maximum charge power on the day, maximum discharge power on the day, charge ampere hour on the day, discharge ampere hour on the day, generating capacity on the day, electricity consumption on the day

It is learned that PDU addresses are 010BH ~ 0114H in turn.

Parsing: in return command

The 4th and 5th bytes indicate minimum voltage of battery on the day:
 $0070H \times 0.1 = 112 \times 0.1 = 11.2V$ (one decimal place)

The 6th and 7th bytes indicate maximum voltage of battery on the day:
 $0084H \times 0.1 = 132 \times 0.1 = 13.2V$ (one decimal place)

The 8th and 9th bytes indicate maximum charge Currentent on the day:
 $00D8H \times 0.01 = 216 \times 0.01 = 2.16A$ (two decimal places)

The 10th and 11th bytes indicate maximum discharge Currentent on the day:
 $006CH \times 0.01 = 108 \times 0.01 = 1.08A$ (two decimal places)

The 12th and 13th bytes indicate maximum charge power on the day: 001CH that is decimalism 28W

The 14th and 15th bytes indicate maximum discharge power on the day: 0010H that is decimalism 16W

The 16th and 17th bytes indicate charging ampere hour on the day: 0608H that is decimalism 1544 AH;

The 18th and 19th bytes indicate discharging ampere hour on the day: 0810H that is decimalism 2064 AH

The 20th and 21st bytes indicate generating capacity on the day: 1388H that is decimalism 5000, then divided by 1000, $5000/1000 = 5kW \cdot h$.

The 22nd and 23rd bytes indicate electricity consumption on the day: 07D0H that is decimalism 2000, then divided by 1000, $2000/1000 = 2 kWh$.

4.2.7 Operation day number of reading, over-discharge times, full discharging times

It is learned that PDU addresses are 0115H, 0116H and 0117H respectively.

Send : 01 03 0115 0003 15F3

Receive : 01 03 06 0008 0001 0006 1176

Parsing:

The 4th and 5th bytes 0008H refer to operation number of days, indicating that the system has already operated for 8 days.

The 6th and 7th bytes 0001H refer to overdischarge times, indicating that the battery has over-discharged at a time.

The 8th and 9th bytes 0006H refer to the full discharge times, indicating that the battery has been fully discharged for 6 times.

4.2.8 Read total charge ampere hour and total discharge ampere hour of battery

It is learned that PDU addresses are 0118H, 0119H, 011AH and 011BH in turn

Send : 01 03 0118 0004 C5F2

Receive : 01 03 08 0001 0203 0000 0108 C0A3

Parsing: The 4th ~ 7th bytes 00010203H refer to total charge ampere hour of battery, decimalism indicates that 66051AH = 66.051KAH

The 8th ~ 11th bytes 00000108H refer to total discharge ampere hour of battery, decimalism indicates that 264AH = 0.264KAH

4.2.9 Read accumulated generating capacity and discharge quantity of controller

It is learned that PDU addresses are 011CH ~ 011FH, totaling 8 bytes

Send : 01 03 011C 0004 8433

Receive : 01 03 08 0000 07D0 0000 03E8 550C

Parsing: the 4th ~ 7th bytes 000007D0H refer to the accumulated generating capacity of the controller, decimalism indicates 2000 (kw·h)

The 8th ~ 11th bytes 000003E8H refer to the accumulated discharge capacity, decimalism indicates 1000(kw·h)

4.2.10 Read street lamp state, brightness and battery state

It is learned that PDU address is 00FDH or 0120H

Each bit in the address indicates the following significance:

PDU address	Byte	Read and write	Item	Data		Meaning
00FDH or 0120H	2	R	State of street lamp	0 or 1	High byte	b7: 0 indicates that the street lamp has been turned off, indicates that the street lamp has been turned on.
			Brightness of	00 ~ 64H		b0 ~ b6: brightness value

			street lamp			
			Battery state		Low byte	00H: no charge is started
				01H: charge mode is started		
				02H: mppt charge mode		
				03H: equalizing charge mode		
				04H: boost charge mode		
				05H: float charge mode		
				06H: Currentent restriction (super power		

Parsing: the 4th byte **E4H** refers to (**80H** | **64H**)

If b7 is 1, indicating the street lamp has been turned on, otherwise it is turned off. b0 ~ b6 is **64H**, indicating that street lamp brightness is 100%

The 5th byte **02H** indicates mppt charge mode (parsing of other state information, please refer to the table above)

4.2.10 Read failure and alarm information

It is learned that PDU addresses are 00FEH ~ 00FFH or 0121H ~ 0122H in turn.

Each bit in the address indicates the following significance:

PDU address	Byte	Read and Write	Item	Byte	Significance of Data
0121H	2	R	Failure alarm information of controller	High 16 bits	B31: open circuit of load (street lamp)
					B30: damage of sensing probe (street lamp)
					B29: over voltage of capacitor (retained)
					B28: reverse connection of battery
					B27: low-temperature protection of battery (the temperature is lower than lower limit of charge) stop charging
					B26: overcharging protection, stop charging
					B25: low-temperature protection of battery (the temperature is lower than the lower limit of discharge) No discharging
					B24: High-temperature protection of battery (the temperature is higher than the upper limit of discharge) No discharging

					B23: no battery is detected (lead acid)
					B22: power supply state (0 indicates power supply from battery, 1 indicates power supply from mains supply)
					B21 ~ B16: retained
0122H	2	R	Failure alarm informa tion of controll er	Low 16 bits	B15 ~ b13: retained
					B12: reverse connection of solar panel
					B11: over voltage of operation point of solar panel
					B10: reverse flowing of solar panel (retained)
					B9: over voltage of photovoltaic input end
					B8: short circuit of photovoltaic input end (retained)
					B7: excessive power at photovoltaic input end
					B6: high-temperature protection of battery (temperature is higher than upper limit of charging) No charging
					B5: excessive temperature of controller
					B4: excessive load power or load over

					Currentent
					B3: short circuit of load
					B2: under voltage alarm
					B1: excessive voltage of battery
					B0: over-discharge of battery

Send : 01 03 0121 0002 95FD

Receive : 01 03 04 8000 0001 1233

Parsing:

The 4th byte is the high 16th bit, with value 8000H, b31 is 1, indicating open circuit of load,

The 5th byte is the low 16th bit, with value 0001H, b0 is 1, indicating overdischarge of battery.

Owing to various failure category, it is unnecessary to parse all, but only the required failure. The common failure inform is recommended as below:

Classific ation of Failure	Type of Failure	Posit ion	Troubleshooting Way	Failure Grade	Affect Normal Operation or Not
Solar battery panel	Inverse connection of solar panel	B12	Inspect whether the wires of positive and negative poles are correctly connected.	Failure	√
	Over voltage of photovoltaic input	B9	Substitute a solar panel with low voltage	Failure	√
	Excessive photovoltaic	B7	The limited power function of the controller may not be	Warning	

	input power		treated.		
Street lamp load	Open circuit of load	B31	Inspect whether load wires are well connected or whether the positive and negative poles of the load are correct	Failure	√
	Excessive load power or load over Currentent	B4	Inspect whether LED load light strings are suitable	Failure	√
	Short circuit of load	B3	Inspect whether load wires are short-circuited.	Failure	√
Battery	Excessive voltage of battery	B1	Inspect whether battery voltage is suitable or whether system voltage is correctly set.	Warning	√
	Over-discharge of accumulator	B0	Inspect whether the battery is fully charged because battery shall restore automatically upon full charge.	Warning	

Note: if the failure grade is defined as “failure” , it is generally required to go to the site for treatment. The failure defined as “warning” grade can be recovered automatically without any treatment or can be restored through simple setting through background.

5 Instruction of Control Command (P3 area)

5.1 Constant current series manual on/off lights with dimming Settings

It is learned that PDU address is DF0AH ~ DF0BH, these two addresses must be set at the same time.

DF0AH refers to manual on/off power with regulation range 0x00 ~ 0x64(decimalism value 0 ~ 100), 0 indicates off.

DF0BH is on/off duration with regulation range 5s ~ 54000s (15 hours).

For example: if it is required to turn on a lamp for 10 minutes with 100% power, following commands are sent:

Command: 01 10 DF0A 0002 04 0064 0258 EEA0

Receive : 01 10 DF0A 0002 5A1E

If it is required to turn on a lamp for 10 minutes with 50% power, following commands are sent:

Command: 01 10 DF0A 0002 04 0032 0258 0EB0

Receive : 01 10 DF0A 0002 5A1E

If it is required to turn off a lamp for 10 minutes, following commands are sent:

Command: 01 10 DF0A 0002 04 0000 0258 AF7F

Receive : 01 10 DF0A 0002 5A1E

After the manual mode is set, the load control can enter into the manual operation mode and the work duration is the time value set manually. After reaching the time set, the controller shall return to automatic work mode.

5.2 Voltage-regulated load series controllers turn on/off the load

In addition to the manual mode with time for the DF0AH ~ DF0BH address, it can also set the value of the DF09 address to change the load mode. This mode can turn on or off the load for a long time, and there is no time limit. If this mode is not saved, the controller would restore to the previous setting of load mode after the power is off.

For example: you need to manually turn on the load, send the command as follows:

Command: 01 06 DF09 0201 A2BC

Receive: 01 06 DF09 0201 A2BC

You need to turn off the load manually, send the command as follows:

Command: 01 06 DF09 0200 637C

Receive: 01 06 DF09 0200 637C

if you need to return to the previously setting load mode, send the command as follows:

Command: 01 06 DF09 0000 621C

Receive: 01 06 DF09 0000 621C

5.3 Reset control of controller

It is learned that PDU address is DF01H, the reset command is as follows:

Command: 01 06 DF01 0001 221E

Receive : 01 06 DF01 0001 221E

After the command is sent, the controller shall restore for restarting.

5.4 Communication module power-off command

After knowing its PDU address is DF0EH, the value indicates how long the delay will be before the communication module is disconnected from the power supply, the unit is second (/s), and the range is 10~54000 seconds. After the controller receives the command, it delays for a period of time (setting value) and then disconnects the power supply of the module for 10 seconds.

The command is as follows:

Command: 01 06 DF0E 0014 D3D2

Receive: 01 06 DF0E 0014 D3D2

Analysis: DF0E is the PDU address, 0014 is the set delay time of 20 seconds.

After sending this command, the controller would delay for 20 seconds. To disconnects the power supply of the communication module for 10 seconds, and then connects again to reset the communication module.

6 Setting Parameter Description of Accumulator (P5 Area)

It is unnecessary to set all battery parameters, but only select the items to be adjusted as per practical demands. Generally, only use the common parameters for setting. During parameter setting, it is recommended to read parameters at first from the device end and then modify the issued setting.

6.1 Parameter setting of lead acid battery

6.1.1 Common parameter of lead acid battery

Lead acid battery contains following common parameters: setting system voltage, battery type, overvoltage, limited charge voltage, equalizing charge voltage, boost charge voltage, float charge voltage, boost charge return voltage, overdischarge return voltage and overdischarge voltage.

It is learned that PDU addresses are E003H ~ E00DH in turn, including 11 words, 22 bytes in total.

For example: it is required to set following parameters (default by sealed lead acid battery)

PDU address	Length	Name	Default	Data Processing	Sending Data	Calculate values to be sent
E003	1	System voltage setting	12V	N*1	12*1=12	000CH
E004	1	Battery type	10 (lead acid)	N*1	10*1=10	000AH
E005	1	Over voltage	15.5V	N*10	17.0*10=170	00AAH
E006	1	Limited charge voltage	15.0V	N*10	15.5*10=155	009BH
E007	1	Equalizing charge voltage	14.6V	N*10	14.6*10=146	0092H

E008	1	Boost charge voltage	14.4V	N*10	14.4*10=144	0090H
E009	1	Float charge voltage	13.8V	N*10	13.8*10=138	008AH
E00A	1	Boost charge return voltage	13.2V	N*10	13.2*10=132	0084H
E00B	1	Over-discharge return voltage	12.6V	N*10	12.6*10=126	007EH
E00C	1	Under voltage alarm voltage	12.0V	N*10	12.0*10=120	0078H
E00D	1	Over-discharge voltage	11.0V	N*10	11.0*10=110	006EH

Send: 01 10 E003 000B 16 000C 000A 00AA 009B 0092 0090 008A 0084 007E 0078 006E 8DCD

Receive: 1 10 E003 000B 460E

6.1.2 Default parameter of different types of lead acid battery

Common lead acid battery contains sealed battery, vented battery and gel battery, the default parameters are as follows:

Battery type	Sealed lead-acid battery SLD	Gel lead-acid battery GEL	Vented lead-acid battery FLD
Over voltage	16.0V	16.0V	16.0V
Charge limit	15.5V	15.5V	15.5V
Equalizing charge voltage	14.6V	—	14.8V

Boost charge voltage	14.4V	14.2V	14.6V
Float charge voltage	13.8V	13.8V	13.8V
Boost charge return voltage	13.2V	13.2V	13.2V
Over-discharge return voltage	12.6V	12.6V	12.6V
Under voltage alarm voltage	12.0V	12.0V	12.0V
Over-discharge voltage	11.0V	11.0V	11.0V

Note: To use 24V system battery, it is only required to set the system voltage as 24V. Parameters of the voltage point are still set based on 12V voltage, the controller shall be multiplied by 2 inside.

6.2 Parameter setting reference of lithium battery

6.2.1 Common setting parameter of lithium battery

Lithium batteries frequently use the parameters: system voltage, battery type, overcharge voltage of lithium battery, overdischarge return voltage, overdischarge return voltage and overdischarge voltage of lithium battery.

Parameters of lithium battery can be set based on following 2 schemes:

(1) E003H ~ E00DH address values are all set at a time, and the unused addresses are set as the default value. Address values E003, E004, E008, E009, E00B and E00D can be sent after modification.

(2) 6 address values E003, E004, E008, E009, E00B and E00D are set independently in different batches. Other address values are not treated.

Use the first plan for setting and it is learned that PDU addresses are E003H ~ E00DH in turn, namely 11 words and 22 bytes in total.

Example: it is required to set parameters in the table below (lithium iron phosphate battery by default)

PDU	Length	Name	Default	Data	Sending	Calculate
------------	---------------	-------------	----------------	-------------	----------------	------------------

Address				Processing	Data	Value To Be Set
E003	1	Setting of system voltage	12V	N*1	12*1=12	000CH
E004	1	Type of battery	11 (Li)	N*1	11*1=11	000BH
E005	1	-	15.5V	N*10	17.0*10=170	00AAH
E006	1	-	15.0V	N*10	15.5*10=155	009BH
E007	1	-	14.6V	N*10	14.6*10=146	0092H
E008	1	Over-discharge voltage of lithium battery	14.4V	N*10	14.4*10=144	0090H
E009	1	Over-discharge return voltage of lithium battery	13.6V	N*10	13.8*10=138	008AH
E00A	1	-	13.2V	N*10	13.2*10=132	0084H
E00B	1	Over-discharge return voltage	12.0V	N*10	12.0*10=120	007EH
E00C	1	-	12.0V	N*10	12.0*10=120	007EH
E00D	1	Over-discharge voltage	10.2V	N*10	10.2*10=102	0066H

Send: 01 10 E003 000B 16 000C 000B 00AA 009B 0092 0090 008A 0084 007E 007E 0066 51F6

Receive: 01 10 E003 000B 460E

6.2.2 Default parameters of different types of lithium battery

Common lithium battery contains lithium iron phosphate and terpolymer lithium battery, the default parameters are as follows:

Default Voltage								
Voltage grade								
Voltage type	Lithium iron phosphate (LiFePO4)	Terpolymer lithium battery (NCM)	Lithium iron phosphate (LiFePO4)	Terpolymer lithium battery (NCM)	Lithium iron phosphate (LiFePO4)	Terpolymer lithium battery (NCM)	Lithium iron phosphate (LiFePO4)	Terpolymer lithium battery (NCM)
Number of batteries in series	1	1	2	2	4	3	8	6
Charge voltage	3.60	4.15	7.2	8.3	14.4	12.5	14.4	12.5
Charge return voltage	3.45	4.00	6.9	8.0	13.8	12.0	13.8	12.0
Over-dischar ge return voltage	3.00	3.60	6.0	7.2	12.0	10.8	12.0	10.8
Over-dischar ge voltage	2.55	3.05	5.1	6.1	10.2	9.2	10.2	9.2

Note: to use 24V system battery, it is only required to set the system voltage as 24V. Parameters of the voltage point are still set based on 12V voltage, the controller shall be multiplied by 2 inside.

7 Description of load setting parameter (P7 area)

Load setting include: load Currentent, intelligent power, light control delay, sensing delay, work time interval. Addresses: 0xE08D ~ 0xE0B3, set uniformly at a time or only set some address data.

7.1 Setting of load Currentent

It is learned that PDU address is E08DH with 0.01 multiplying power. The value can be set only after being multiplied by 0.01. For example, to set 0.9A Currentent, it is required to send $0.9 \times 100 = 90$ Currentent practically, which is 5A after being converted to hexadecimal.

Currentent parameters are set through following commands:

Send : 01 06 E08D 005A AE1A

Receive : 01 06 E08D 005A AE1A

7.2 Setting of intelligent power

Intelligent power can be achieved through setting on/off or selecting different levels, so as to realize different energy-saving effect. The corresponding energy-saving effects of intelligent power are shown in the table below.

E08E address value	Level of intelligent power	Energy-saving effect	Load brightness when the battery runs out	Lighting duration
0	Off	Without energy-saving effect, it only controls load power with time interval.	Brightest	Shortest
1	High	Best energy-saving effect	Darkest	Longest
2	Middle	Better energy-saving effect	Bright	Long
3	Low	Poor energy-saving effect	Brighter	Shorter
4	Automatic	Automatically select one level from "high", "middle" and "low" levels for operation	Intelligent learning Automatic adjustment	Intelligent learning Automatic adjustment
5	User-defined	Users independently define voltage and minimum Currentent value at energy-saving start point and end point.	Confirmed based on user setting.	Confirmed based on user setting.

It is recommended to use "middle" in intelligent powder level to meet most applications.

It is learned that PDU address is E08EH, the "middle" commands set in intelligent power are as following:

Send : 01 06 E08E 0002 5FE0

Receive : 01 06 E08E 0002 5FE0

7.3 Light control delay setting

Light control delay is generally set as 5 seconds~10 minutes. It is recommended to use 20s to meet most applications.

It is learned that PDU address is E08FH and 20s light control delay is set with the commands below:

Send : 01 06 E08F 0014 8FEE

Receive : 01 06 E08F 0014 8FEE

7.4 Sensing delay setting

Sensing delay duration shall be set for products with human body sensing. For products without human body sensing, the sensing delay duration shall be set as 0. When sensing delay is not set as 0, unattended power in time interval control can be enabled automatically. Otherwise, there is only attended power enabling in the time interval control.

It is learned that PDU address is E090H. The sensing delay shall be set as 0 for the products without sensing function with the following commands:

Send : 01 06 E090 0000 BE27

Receive : 01 06 E090 0000 BE27

For products with sensing function, the sensing delay duration is generally set as 20s with the following commands:

Send : 01 06 E090 0014 BE28

Receive : 01 06 E090 0014 BE28

7.5 Setting of operation duration and power

10time intervals are supported in operation duration at most (9time intervals+ Morninging on). some or all time intervals can be used as per required. It is required to note that when it is required to use unattended power, the "sensing delay" value shall be set as a value other than 0. When no unattended power is required, the unattended power address value can be set as 0.

7.5.1 Senseless mode setting

For example: 5time interval controllers but no unattended power is required.

Time interval	Duration	Power
The 1 st time interval	0.5h	50%
The 2 nd time interval	1h	100%
The 3 rd time interval	2h	70%
The 4 th time interval	3h	50%
The 5 th time interval	5h	30%

Corresponding PDU address value:

PDU address	Length	Name	Set value	Data processing	Sent data	Calculate value to be sent
E092	1	Light on duration of first section	0.5h	N*3600	$0.5 \times 3600 = 1800$	0708H
E093	1	Attended power of first section	50%	N*1	$50 \times 1 = 50$	0032H
E094	1	Unattended power of first section	0%	N*1	$0 \times 1 = 0$	0000H
E095	1	Light on duration of second section	1h	N*3600	$1 \times 3600 = 3600$	0E10H
E096	1	Attended power of second section	100%	N*1	$100 \times 1 = 100$	0064H
E097	1	Unattended power of second section	0%	N*1	$0 \times 1 = 0$	0000H

E098	1	Light on duration of third section	2h	N*3600	$2*3600 = 7200$	1C20H
E099	1	Attended power of third section	70%	N*1	$70*1 = 70$	0046H
E09A	1	Unattended power of third section	0%	N*1	$0*1 = 0$	0000H
E09B	1	Light on duration of fourth section	3h	N*3600	$3*3600 = 10800$	2A30H
E09C	1	Attended power of fourth section	50%	N*1	$50*1 = 50$	0032H
E09D	1	Unattended power of fourth section	0%	N*1	$0*1 = 0$	0000H
E09E	1	Light on duration of fifth section	5h	N*3600	$5*3600 = 18000$	4650H
E09F	1	Attended power of fifth section	30%	N*1	$30*1 = 30$	001EH
E0A0	1	Unattended power of fifth section	0%	N*1	$0*1 = 0$	0000H

Send the following commands:

Send: 01 10 E092 000F 1E 0708 0032 0000 0E10 0064 0000 1C20 0046 0000 2A30 0032 0000 4650 001E 0000 13B0

Receive: 01 10 E092 000F 1620

7.5.2 Sensing mode setting

For example: 5time interval controllers and setting of unattended power are required.

Time Interval	Duration	Attended Power	Unattended Power
The 1 st time interval	0.5h	50%	30%
The 2 nd time interval	1h	100%	70%
The 3 rd time interval	2h	70%	50%
The 4 th time interval	3h	50%	30%
The 5 th time interval	5h	30%	10%

Corresponding PDU address value:

PDU address	Length	Name	Set value	Data processing	Sent data	Calculate value to be sent
E092	1	Light on duration of first section	0.5h	N*3600	$0.5 \times 3600 = 1800$	0708H
E093	1	Attended power of first section	50%	N*1	$50 \times 1 = 50$	0032H
E094	1	Unattended power of first section	30%	N*1	$30 \times 1 = 30$	001EH
E095	1	Light on duration of second section	1h	N*3600	$1 \times 3600 = 3600$	0E10H
E096	1	Attended power of second section	100%	N*1	$100 \times 1 = 100$	0064H

E097	1	Unattended power of second section	70%	N*1	$70*1 = 70$	0046H
E098	1	Light on duration of third section	2h	N*3600	$2*3600 = 7200$	1C20H
E099	1	Attended power of third section	70%	N*1	$70*1 = 70$	0046H
E09A	1	Unattended power of third section	50%	N*1	$50*1 = 50$	0032H
E09B	1	Light on duration of fourth section	3h	N*3600	$3*3600 = 10800$	2A30H
E09C	1	Attended power of fourth section	50%	N*1	$50*1 = 50$	0032H
E09D	1	Unattended power of fourth section	30%	N*1	$30*1 = 30$	001EH
E09E	1	Light on duration of fifth section	5h	N*3600	$5*3600 = 18000$	4650H
E09F	1	Attended power of fifth section	30%	N*1	$30*1 = 30$	001EH
E0A0	1	Unattended	10%	N*1	$10*1 = 0$	000AH

		power of fifth section				
--	--	---------------------------	--	--	--	--

Send the following commands:

**Send:01 10 E092 000F 1E 0708 0032 001E 0E10 0064 0046 1C20 0046 0032 2A30 0032
001E 4650 001E 000A B60B**

Receive:01 10 E092 000F 1620